

SMART INDIA HACKATHON 2026



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HACKATHON
2026

VETNEXA

- **Problem Statement ID** – SIH26128
- **Problem Statement Title** - Efficient system for early detection, prevention and management of livestock diseases and animal health issues.
- **Theme** - MedTech / BioTech / HealthTech
- **PS Category** - Software
- **Team ID-** 192212
- **Team Name** - **VETNEXA**



PROPOSED SOLUTION

VETNEXA is an AI-powered platform for early detection, prevention and management of livestock diseases and animal health issues. It combines symptoms, animals images, health history and data-driven insights to support farmers and veterinarians.

HOW IT ADDRESSES THE PROBLEM



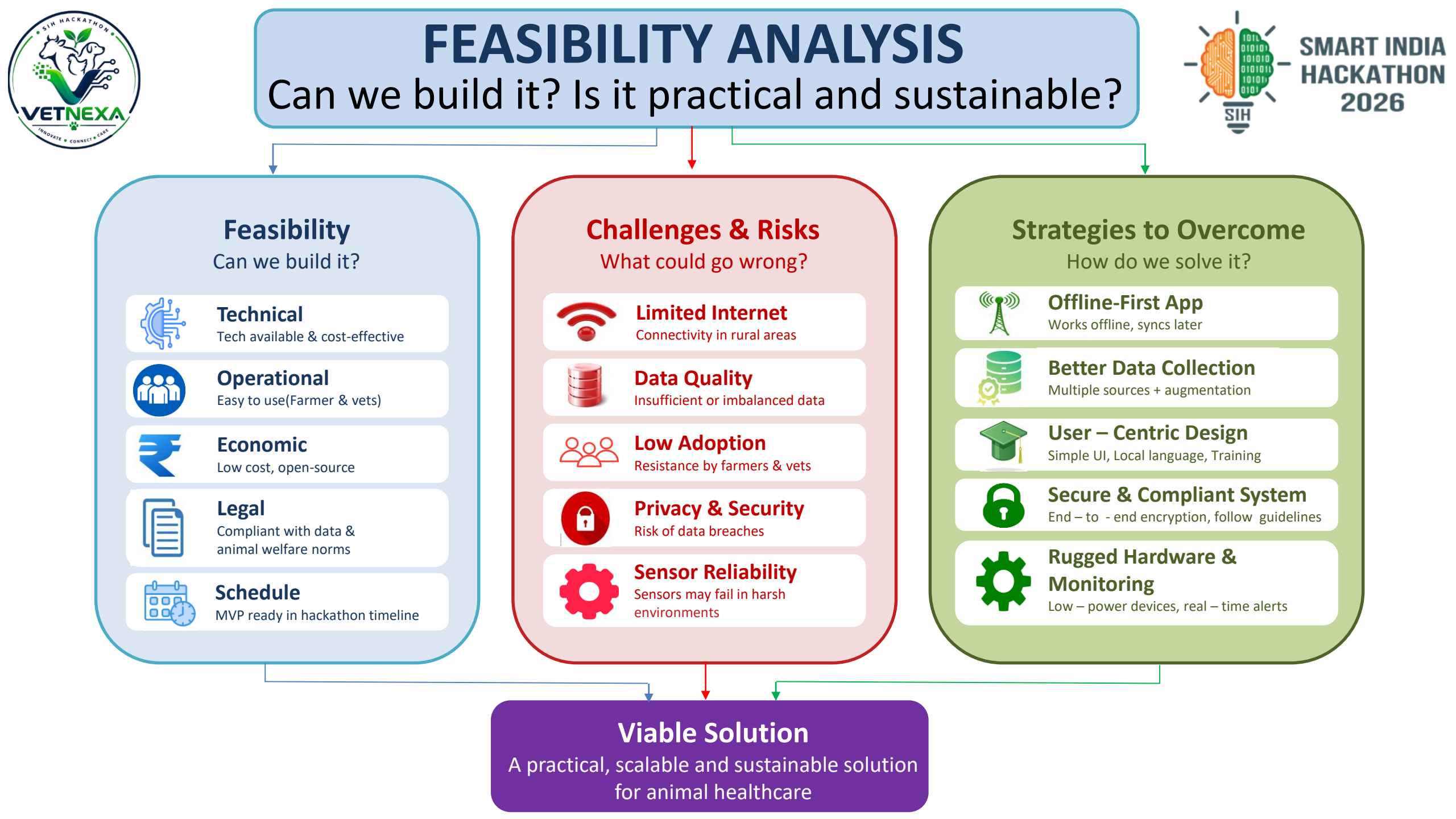
INNOVATION AND UNIQUENESS OF THE SOLUTION

- Multi-Model AI** → Symptoms + Image + History for a richer assessment.
- Individual --> Herd Intelligence** → Animals-level data can reveal herd patterns and early outbreak signals.
- Farmer + Veterinarians Workflow** → AI-Assisted assessment connects to consultation and follow-up.
- Regional Language Assistant** → Voice/Chat interaction for easier farmer access.

OUR SOLUTION AT A GLANCE



Healthier Animals, Stronger Farmers, Sustainable Future



TECHNOLOGY TO BE USED

	Programming Language HTML5, CSS3, JavaScript
	Frontend React.js (Web Interface)
	Backend Node.js (Server – Side application)
	API Express.js (REST API and communication)
	Database MongoDB (Animal profile, health records, cases)
	AI/ML Python (AI/ML development)
	Deep Learning TensorFlow / Py-Torch (Disease prediction model)
	Image Processing OpenCV (Image preprocessing and analysis)
	AI Model CNN / Transfer Learning (Livestock disease classification)
	Authentication Firebase Authentication (Secure farmer / veterinarian login)
	Storage Firebase Storage / Cloud Storage (Store uploaded animal images)
	Deployment Firebase / AWS / Render (Hosting and deployment)
	Camera WebRTC / get User Media (Real – time image capture)
	Notification - Firebase Cloud Messaging (Health alert and notifications)



TARGET AUDIENCE



Farmers & Diary Owners

Get early alerts, guidance and better herd management



Veterinarians

Quick diagnosis support, case tracking and consultation.



Government & NGOs

Monitor disease outbreaks and plan effective interventions.



Livestock Businesses

Reduce losses, improve productivity and ensure better animal welfare.

POTENTIAL IMPACT ON THE TARGET AUDIENCE



EARLY DISEASE DETECTION

Detect diseases early and take timely action, reducing spread and mortality.



INCREASED PRODUCTIVITY

Healthy animals lead to higher milk production, growth and performance.



BETTER ANIMAL HEALTH & WELFARE

Improves overall health, productivity and quality of life for animals.



REDUCED ECONOMIC LOSSES

Prevents outbreaks. reduces treatment costs and minimizes financial losses.



INFORMED DECISION MAKING

Data - driven insights help in better planning and resource management.



REAL TIME ALERTS & RISK MITIGATION

Instant alerts and risk mapping help in taking preventive measures and control outbreaks.

BENEFITS OF THE SOLUTION



SOCIAL BENEFIT

Better animal health ensures food security and improves rural livelihood.



ECONOMIC BENEFIT

Increased productivity and reduced losses lead to higher income for farmers and businesses.



ENVIRONMENTAL BENEFIT

Promotes sustainable livestock management and reduces unnecessary use of medicines



NATIONAL BENEFIT

Strength livestock sector, contributes to rural development and supports Aatmnirbhar Bharat.



RESEARCH AND REFERENCES

Knowledge. Innovation. Impact



RESEARCH BACKGROUND

Livestock diseases cause major economic losses and threaten food security. Early detection, continuous monitoring and timely interventions can significantly reduce losses and improve animal productivity. Our solution leverages AI, IoT and data analytics to provide intelligent health management for livestock.

REFERENCES AND RESEARCH WORK

Food and Agriculture Organization (FAO) - Animal Health

<https://openknowledge.fao.org/server/api/core/bitstreams/9acc7045-780e-4819-8a1e-f999ca4f7107/content>

OIE - World Organisation for Animals Health

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